

ch27 Nuclear Energy

9 item(s)

Answers written in the margins will not be marked.

11. Radium-226 ($^{226}_{88}\text{Ra}$) undergoes α -decay into radon (Rn).

(a) Write a nuclear equation for the decay.

(2 marks)

* (b) Given : mass of a radium nucleus = 226.0254 u

mass of a radon nucleus = 222.0176 u

mass of an α -particle = 4.0026 u

Calculate the energy released in the decay in MeV.

(2 marks)

(c) 1 curie (Ci) is defined as the activity of 1 g of radium. The activity of a radium source used in laboratories is about 5 μCi . Estimate the number of radium atoms in this source and hence find its activity expressed in disintegrations per second. The half-life of radium-226 is 1600 years and take the mass of one mole of radium as 226 g. ($1 \mu\text{Ci} = 1 \times 10^{-6} \text{ Ci}$)

(3 marks)

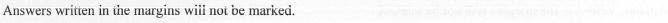
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(2 marks)

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(1 mark)

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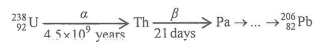
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(2 marks)

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9. Part of the decay series of uranium-238 (U-238) is shown below. The end product lead-206 (Pb-206) is stable.



- (a) When a U-238 nucleus decays to a Pb-206 nucleus, how many α -particle(s) and β -particle(s) are emitted? (2 marks)

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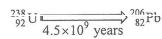
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- (b) As the first decay in the above chain from U to Th has a half-life much longer than those of subsequent decays, the decay from U-238 to Pb-206 can be simplified to a *single decay* with half-life 4.5×10^9 years:



Suppose that a uranium-bearing rock contains only U-238 and no Pb-206 at the time when it was formed long ago by solidification of molten material. In a particular sample of the rock, it is found that the

$$\frac{\text{number of Pb-206 atoms}}{\text{number of U-238 atoms}} = \frac{2}{3} \text{ at present}$$

- (i) Estimate the age of the rock. Assume that all Pb-206 atoms come from the decay of U-238 originally present in the sample and ignore the small number of U-238 atoms which have decayed but have not yet become Pb-206. (2 marks)

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- (ii) State, with a reason, whether the answer in (b)(i) is an overestimate or an underestimate of the age of the rock if some Pb-206 atoms have actually been lost. (2 marks)

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- (iii) The graph in Figure 9.1 shows how the number of U-238 atoms in the sample varies with time t subsequently while $t=0$ denotes the *present time*. On Figure 9.1, sketch a graph to show the variation of the number of Pb-206 atoms in the sample with time. (2 marks)

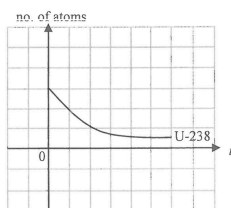


Figure 9.1

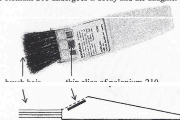
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10. Dust may adhere to the surfaces of photos and films due to electrostatic attraction. To remove the dust effectively, a special brush with a thin slice of polonium-210 ($^{210}_{84}\text{Po}$) fixed near the brush hair as shown in Figure 10.1 may be used. Polonium-210 undergoes α decay and the daughter nucleus lead (Pb) is stable.

Figure 10.1



(a) Write a nuclear equation for the decay of polonium-210. (2 marks)

(b) Briefly explain how the α particles help clean the charged dust. (2 marks)

(c) Briefly explain why the polonium-210 slice must be fixed near to the brush hair. (1 mark)

* (d) The manufacturer recommends that the brush should be returned to the factory for replacement of the polonium-210 slice every year. Taking the activity of a newly replaced polonium-210 slice as 1 unit, find its activity after one year (365 days). Given: half-life of polonium-210 is 138 days. (2 marks)

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List of data, formulae and relationships

Data

molar gas constant	$R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$
Avogadro constant	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
acceleration due to gravity	$g = 9.81 \text{ m s}^{-2}$ (close to the Earth)
universal gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
speed of light in vacuum	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
charge of electron	$e = 1.60 \times 10^{-19} \text{ C}$
electron rest mass	$m_e = 9.11 \times 10^{-31} \text{ kg}$
permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$
permeability of free space	$\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$
atomic mass unit	$u = 1.661 \times 10^{-27} \text{ kg}$ (1 u is equivalent to 931 MeV)
astronomical unit	$\text{AU} = 1.50 \times 10^8 \text{ m}$
light year	$\text{ly} = 9.46 \times 10^{15} \text{ m}$
parsec	$\text{pc} = 3.09 \times 10^{16} \text{ m} = 3.26 \text{ ly} = 206265 \text{ AU}$
Stefan constant	$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
Planck constant	$h = 6.63 \times 10^{-34} \text{ J s}$

Mathematics

For uniformly accelerated motion:	Equation of a straight line	$y = mx + c$
$v = u + at$	Area of triangle	$\frac{1}{2}bh$
$s = ut + \frac{1}{2}at^2$	Surface area of cylinder	$2\pi rh + 2\pi r^2$
$v^2 = u^2 + 2as$	Volume of cylinder	$\pi r^2 h$
	Surface area of sphere	$4\pi r^2$
	Volume of sphere	$\frac{4}{3}\pi r^3$
	For small angles, $\sin \theta \approx \tan \theta \approx \theta$ (in radians)	

Astronomy and Space Science	Energy and Use of Energy
$U = -\frac{GMm}{r}$ gravitational potential energy	$E = \frac{Q}{d}$ illuminance
$P = \sigma AT^4$ Stefan's law	$\frac{Q}{t} = \frac{A(T_1 - T_2)}{d}$ rate of energy transfer by conduction
$\frac{M_1}{r_1^2} = \frac{M_2}{r_2^2}$	$U = \frac{Q}{d}$ thermal transmittance U-value
	$P = \frac{1}{2} \rho A v^3$ maximum power by wind turbine
Atomic World	Medical Physics
$\frac{1}{2} m_e v_{\text{max}}^2 = hf - \phi$ Einstein's photoelectric equation	$\theta = \frac{1.22\lambda}{d}$ Rayleigh criterion (resolving power)
$E_n = \frac{1}{2} \left(\frac{m_e e^4}{4\pi\epsilon_0^2 h^2} \right) \frac{1}{n^2}$ 13.6 eV	$\text{power} = \frac{1}{f}$ power of a lens
energy level equation for hydrogen atom	$L = 10 \log \frac{I}{I_0}$ intensity level (dB)
$\lambda = \frac{h}{p}$ de Broglie formula	$Z = \rho c$ acoustic impedance
$\theta = \frac{1.22\lambda}{d}$ Rayleigh criterion (resolving power)	$\alpha = \frac{I_r}{I_0} = \frac{(Z_2 - Z_1)^2}{(Z_2 + Z_1)^2}$ intensity reflection coefficient
	$I = I_0 e^{-\mu x}$ transmitted intensity through a medium

A1. $E = mc \Delta T$ energy transfer during heating and cooling	D1. $F = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2}$ Coulomb's law
A2. $Q = I \Delta t$ energy transfer during charge at rates	D2. $E = \frac{Q}{4\pi\epsilon_0 r^2}$ electric field strength due to a point charge
A3. $pV = nRT$ equation of state for an ideal gas	D3. $E = \frac{V}{u}$ electric field between parallel plates
A4. $pV = \frac{1}{3} Nmc^2$ kinetic theory equation	D4. $R = \frac{\rho l}{A}$ resistance and resistivity
A5. $E_k = \frac{2RT}{2N_A}$ molecular kinetic energy	D5. $R = R_1 + R_2$ resistors in series
B1. $F = m \frac{\Delta v}{\Delta t} = \frac{\Delta p}{\Delta t}$ force	D6. $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$ resistance in parallel
B2. $\text{moment} = F \times d$ moment of a force	D7. $F = IF = I \times B$ forces on a current-carrying wire in a magnetic field
B3. $E_p = mgh$ gravitational potential energy	D8. $F = BIl \sin \theta$ forces on a current-carrying conductor in a magnetic field
B4. $E_k = \frac{1}{2} mv^2$ kinetic energy	D9. $F = BIl \sin \theta$ forces on a current-carrying conductor in a magnetic field
B5. $P = IV$ mechanical power	D10. $B = \frac{\mu_0 I}{2\pi r}$ magnetic field due to a long straight wire
B6. $a = \frac{v^2}{r} = \omega^2 r$ centripetal acceleration	D11. $B = \frac{\mu_0 NI}{2\pi r}$ magnetic field due to a long solenoid
B7. $F = \frac{Gm_1 m_2}{r^2}$ Newton's law of gravitation	D12. $\mathcal{E} = N \frac{\Delta \Phi}{\Delta t}$ induced e.m.f.
C1. $\lambda_N = \frac{h}{p}$ de Broglie formula	D13. $\frac{V_s}{N_s} = \frac{V_p}{N_p}$ ratio of secondary voltage to primary voltage in a transformer
C2. $d \sin \theta = n\lambda$ diffraction grating equation	E1. $N = N_0 e^{-\lambda t}$ law of radioactive decay
C3. $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ equation for a single lens	E2. $t_{1/2} = \frac{\ln 2}{\lambda}$ half-life and decay constant
	E3. $A = \lambda N$ activity and the number of nuclei
	E4. $\lambda = \frac{h}{p}$ de Broglie formula

10. (a) The equation below represents nuclear fission of uranium-235 (U-235).

$${}_{92}^{235}\text{U} + {}_0^1\text{n} \rightarrow {}_{56}^{141}\text{Ba} + {}_{36}^{92}\text{Kr} + x{}_0^1\text{n} + 200\text{ MeV}$$

(i) What is the value of x ? (1 mark)

(ii) State a necessary condition for chain reaction of fission to occur. (1 mark)

Scientists found evidence in Oklo, Africa that natural nuclear fission occurred two billion (2×10^9) years ago. The uranium mineral ore mined from Oklo at present is found to have 0.6% concentration by mass of U-235 (see the table below), which is much lower than usual.

(b) The table gives the information of U-235 and U-238 in a sample of uranium mineral ore found in Oklo. Given: half-life of U-235 = 7.04×10^8 years

	2×10^9 years ago	at present
U-235	m_0 kg	0.060 kg (i.e. 0.6% concentration by mass)
U-238	13.556 kg	9.940 kg (i.e. 99.4% concentration by mass)

*(i) Estimate the amount m_0 (in kg) of U-235 in the sample 2×10^9 years ago. (2 marks)

(ii) Hence determine whether natural nuclear fission of U-235 was possible 2×10^9 years ago. For fission of U-235 to happen, its concentration by mass in the uranium mineral ore has to be at least 3%. (1 mark)

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There must be underground water in the vicinity of this uranium-rich mineral deposit for natural nuclear fission to be possible. Since water can slow down the fast neutrons from fission, these neutrons can easily be captured by U-235.

(c) In fact the chain reaction stopped even before the concentration by mass of U-235 dropped to 3%. Explain why this occurred. (2 marks)

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10. Given: mass of proton = 1.0073 u
 mass of neutron = 1.0087 u
 mass of $^{14}_7\text{N}$ nucleus = 13.9992 u
 mass of $^{17}_8\text{O}$ nucleus = 16.9947 u

When a stationary $^{14}_7\text{N}$ nucleus is bombarded by an α particle, the following nuclear reaction can be triggered with products $^{17}_8\text{O}$ and X fly off:



- (a) What is X ? (1 mark)

*(b)Based on energy consideration, estimate the minimum kinetic energy, in MeV, of the α particle required for such a nuclear reaction to occur. (2 marks)

- (c) However, when conservation of momentum is also taken into account, the α particle must possess a kinetic energy greater than that found in (b) to bring about such a reaction. Explain. (2 marks)

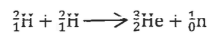
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10. Deuterium (${}^2_1\text{H}$) and tritium (${}^3_1\text{H}$) are isotopes of hydrogen. The fusion of two deuterium nuclides can be represented by the following equation:



Given: mass of ${}^2_1\text{H} = 2.014102 \text{ u}$
 mass of ${}^3_2\text{He} = 3.016029 \text{ u}$
 mass of ${}^1_0\text{n} = 1.008665 \text{ u}$

- *(a) 1 in 6420 atoms of hydrogen in nature is a deuterium. Estimate the maximum energy, in MeV, that could be produced from this fusion reaction with 1 mole of hydrogen nuclides. (3 marks)

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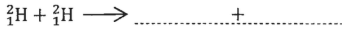
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- (b) If conditions are altered, the fusion of two deuterium nuclides may **not** produce a helium (${}^4_2\text{He}$) nucleus. Complete the equation below for such a possible fusion reaction. (1 mark)



- (c) Both fission and fusion produce energy. State **TWO** advantages of using fusion as an energy source over fission. (2 marks)

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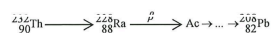
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9. Part of the decay series of thorium-232 (Th-232) is shown below. Th-232 decays to radium-228 (Ra-228) with a half-life of 1.41×10^{10} years. The end product lead-208 (Pb-208) is stable.



- (a) (i) State the kind of radiation emitted by Th-232. (1 mark)

- (ii) How many α - and β -decay(s) has occurred before a Th-232 nucleus becomes a Pb-208 nucleus? (2 marks)

- *(b) Estimate the proportion of Th-232 decaying in 10 years. Give your answer correct to 2 significant figures. (2 marks)

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- (c) By adding thorium oxide (a compound containing Th-232) to glass, thoriated glass having a higher refractive index is produced. Camera lenses manufactured before 1970 are often made from thoriated glass.

Figure 9.1



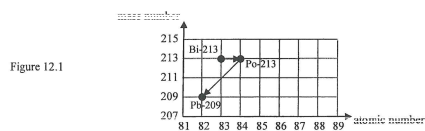
- (i) Although thoriated glass is radioactive, practically it is safe to use a camera with such lenses like the one shown in Figure 9.1. Why? (1 mark)

- (ii) According to the decay series of thorium-232, thorium oxide would end up as lead oxide which is black in colour. Why is there no need to worry that a thoriated glass lens would turn black when used for several years? (1 mark)

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12. (a) Bismuth-213 (Bi-213) is a radionuclide currently undergoing clinical trials for cancer treatment. It is prepared in liquid and injected into the human body, where it targets and destroys cancer cells. Figure 12.1 shows a decay series of Bi-213 .



- (i) Write the nuclear equations corresponding to each of these two decay processes. (2 marks)

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- *(ii) The half-life of Bi-213 is 46 minutes. At time $t = 0$ s, its number of undecayed nuclei is 8.0×10^{11} . Find the activity (A in Bq) at $t = 0$ s. (2 marks)

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- *(iii) Find the percentage decrease in the activity after 200 minutes. (2 marks)

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- (b) In hospital, technicians wear protective gloves to handle liquid radioactive materials, such as those containing Bi-213 . To monitor radiation exposure, they wear a specialised ring on their finger, as shown in Figure 12.2(a). Each ring contains a small chip in a plastic enclosure (Figure 12.2(b)), which detects and records radiation doses. The rings are collected every three months to monitor the radiation exposure of the technicians.



Figure 12.2(a)

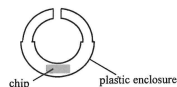


Figure 12.2(b)

- (i) Compared to placing it in a badge worn on the chest, what is the advantage of placing the chip in the ring worn on the hand? (1 mark)

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- (ii) The ring should be worn under the gloves. Explain briefly. (1 mark)

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- (iii) Which type(s) of radiation (α , β or γ) can reach the chip of the ring? (1 mark)

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